

SCQ (Single Correct Type) :

- $$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\sin \frac{\pi}{2k} - \cos \frac{\pi}{2k} - \sin \left(\frac{\pi}{2(k+2)} \right) + \cos \frac{\pi}{2(k+2)} \right) =$$

(A) 0 (B) 1 (C) 2 (D) 3
- The number of critical points of $f(x) = \left(\int_0^x (\cos^2 t - \sqrt[3]{t}) dt \right) + \frac{3}{4}x^{4/3} - \frac{x+1}{2}$ in $[0, 6\pi]$ is :

(A) 10 (B) 8 (C) 6 (D) 12
- Evaluate

$$\int \frac{\left(\sqrt[3]{x + \sqrt{2-x^2}} \right) \left(\sqrt[6]{1-x\sqrt{2-x^2}} \right) dx}{\sqrt[3]{1-x^2}}; x \in (0,1):$$

(A) $2^{1/6}x + C$ (B) $2^{1/12}x + C$ (C) $2^{1/3}x + C$ (D) None of these
- The solution of the differential equation $\sin 2y \frac{dy}{dx} + 2 \tan x \cos^2 y - 2 \sec x \cos^3 y$ is :

(where C is arbitrary constant)

(A) $\cos y \sec x = \tan x + C$ (B) $\sec y \cos x = \tan x + C$
 (C) $\sec y \sec x = \tan x + C$ (D) $\tan y \sec x = \sec x + C$
- Solution of the differential equation $(2 + 2x^2\sqrt{y})ydx + (x^2\sqrt{y} + 2)x dy = 0$ is/are :

(A) $xy(x^2\sqrt{y} + 5) = c$ (B) $xy(x^2\sqrt{y} + 3) = c$ (C) $xy(y^2\sqrt{x} + 3) = c$ (D) $xy(y^2\sqrt{y} + 5) = c$
- Let $A \equiv (-2, 0)$ and $B \equiv (2, 0)$, then the number of integral values of a , $a \in [-10, 10]$ for which line segment AB subtends an acute angle at point $C(a, a+1)$ is

(A) 15 (B) 17 (C) 19 (D) 21

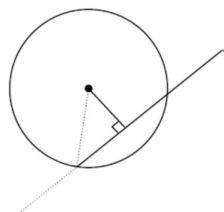
MCQ (One or more than one correct) :

- For $a > 0$, if $I = \int \sqrt{\frac{x}{a^3 - x^3}} dx = A \sin^{-1} \left(\frac{x^{3/2}}{B} \right) + C$, where C is any arbitrary constant, then:

(A) $A = \frac{2}{3}$ (B) $B = a^{3/2}$ (C) $A = \frac{1}{3}$ (D) $B = a^{1/2}$

8. Let $\int x \sin x \cdot \sec^3 x dx = \frac{1}{2}(x \cdot f(x) - g(x)) + k$, then:
- (A) $f(x) \notin (-1, 1)$ (B) $g(x) = \sin x$ has 6 solution for $e \in [-\pi, 2\pi]$
- (C) $g'(x) = f(x), \forall x \in \mathbb{R}$ (D) $f(x) = g(x)$ has no solution
9. $\int \sqrt{\tan^2 x + 2} dx = a \tan^{-1}\left(\frac{\tan x}{\sqrt{2 + \tan^2 x}}\right) + C$, where C is constant of integration, then
- (A) $a = 1$ (B) $a = 2$
- (C) Range of $f(x)$ is $[2, \infty)$ (D) Range of $f(x)$ is $[3, \infty)$
10. Solution of the differential equation $\frac{dy}{dx} = \frac{y^2 - 2xy - x^2}{y^2 + 2xy - x^2}$
- (A) $x^2 - y^2 + c(x - y) = 0$ (B) $x^2 + y^2 + c(x + y) = 0$
- (C) a straight line if it passes through $(1, -1)$ (D) a circle if it passes through $(1, 1)$
11. If $y(x)$ satisfies the differential equation $\frac{dy}{dx} = \sin 2x + 3y \cot x$ and $y\left(\frac{\pi}{2}\right) = 2$ then which of the following statement(s) is/are correct?
- (A) $y\left(\frac{\pi}{6}\right) = 0$ (B) $y'\left(\frac{\pi}{3}\right) = \frac{9 - 3\sqrt{2}}{2}$
- (C) $y(x)$ increases in the interval $\left(\frac{\pi}{6}, \frac{\pi}{2}\right)$ (D) $\int_{-\pi/2}^{\pi/2} y(x) dx = x$
12. C_1 and C_2 are two circles of radii a and b ($a < b$) touching both the coordinate axes and have their centres in the first quadrant. Then which of the following is true?
- (A) If C_1, C_2 touch each other then $\frac{b}{a} = 3 + 2\sqrt{2}$
- (B) If C_1, C_2 are orthogonal then $\frac{b}{a} = 2 + \sqrt{3}$
- (C) If C_1, C_2 intersect in such a way that their common chord has maximum length, then $\frac{b}{a} = 3$
- (D) If C_2 passes through the centre of C_1 , then $\frac{b}{a} = 2 + \sqrt{2}$

13. From point $P(a, b)$ a line, L , is drawn to meet $\alpha x^2 + 2hxy + \beta y^2 = 1$ at Q and R . If $PQ \cdot PR$ is independent of slope of line L , then :



- (A) $\alpha = 1 \Rightarrow \beta = 1$ (B) $\alpha = 2 \Rightarrow h = 0$
 (C) $h = 0 \Rightarrow \beta = 2$ (D) If $\alpha \in \mathbb{N}$, then $\alpha\beta$ is a perfect square
14. Circle, $C_1 : x^2 + y^2 - 8x - 6y + 7 = 0$ intersects the circle $C_2 : x^2 + y^2 - 2ax + b = 0$ orthogonally. Given that C_2 is a real circle of non-zero radius and $a, b, x, y \in \mathbb{R}$
 (A) $8a - b = 9$ (B) $8a - b = 7$
 (C) If a is prime natural number $a_{\min} = 7$ (D) If a is prime natural number $a_{\min} = 11$
15. For the circles $x^2 + y^2 - 10x + 16y + 89 - r^2 = 0$ and $x^2 + y^2 + 6x - 14y + 42$ which of the following is/are true.
 (A) Number of integral values of r are 7 for which circles are intersecting.
 (B) Number of integral values of r are 9 for which circles are intersecting.
 (C) For r equal to 13 number of common tangents are 3.
 (D) For r equal to 21 number of common tangents are 2.

Numerical Based Questions :

16. If $f(x)$ is continuous and differentiable in $[-3, 9]$ and $f'(x) \in [-2, 8] \forall x \in (-3, 9)$. Let N be the number of divisors of the greatest possible value of $f(9) - f(-3)$, then find the sum of digits of N .
17. Let $f(x) = ax + \cos 2x + \sin x + \cos x$ is defined for $\forall x \in \mathbb{R}$ and $a \in \mathbb{R}$ and is strictly increasing function. If the range of a is $\left[\frac{m}{n}, \infty\right)$, then find the minimum value of $(m - n)$.
18. Let $y = f(x)$ satisfies the differential equation $xy(1 + y)dx = dy$. If $f(0) = 1$ and $f(2) = \frac{e^2}{k - e^2}$, then find the value of k .

Subjective Type Problems

19. The plane denoted by $\Pi_1 : 4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line of intersection with the plane $\Pi_2 : 5x + 3y + 10z = 25$. If the plane in its new position be denoted by n , and the distance of this plane from the origin is $\sqrt{53}k$ where $k \in \mathbb{N}$, then find k .



ANSWER KEY OF CAPS – 14+

1.	(D)	2.	(D)	3.	(A)	4.	(C)	5.	(B)	6.	(C)	7.	(AB)
8.	(ACD)	9.	(AC)	10.	(BCD)	11.	(AC)	12.	(ABCD)				
13.	(ABD)	14.	(BD)	15.	(AC)	16.	3	17.	9	18.	2		